GENED 1079 Section 5

Week of October 13, 2026

Agenda

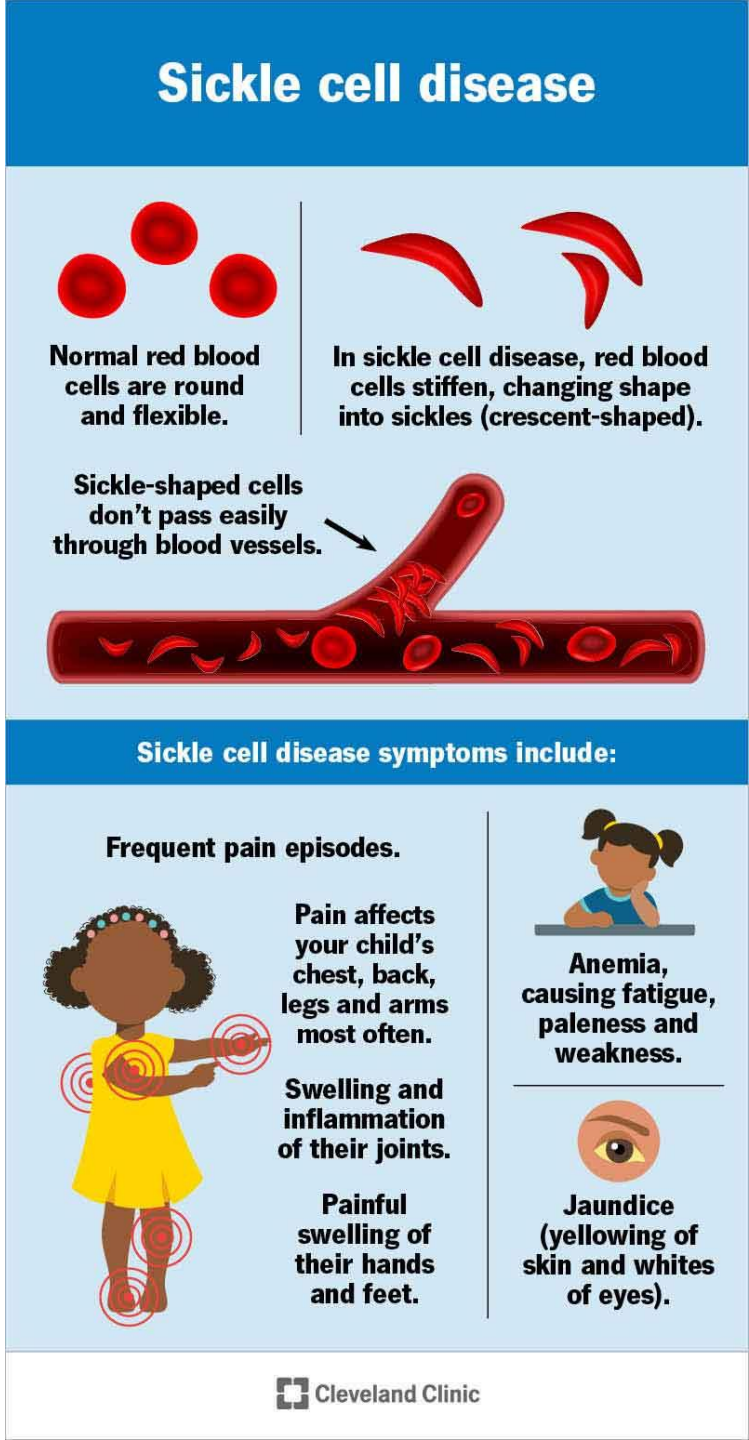
1. Intro: Sickle Cell Disease and Cell and Gene Therapies *(15 min)*
2. Group Exercise: Decide how your state will allocate gene therapies for people with Medicaid *(20 min)*
3. Presentations & discussion *(25 min)*
4. Access to therapies in low-resource settings / low and middle-income countries *(15 min)*

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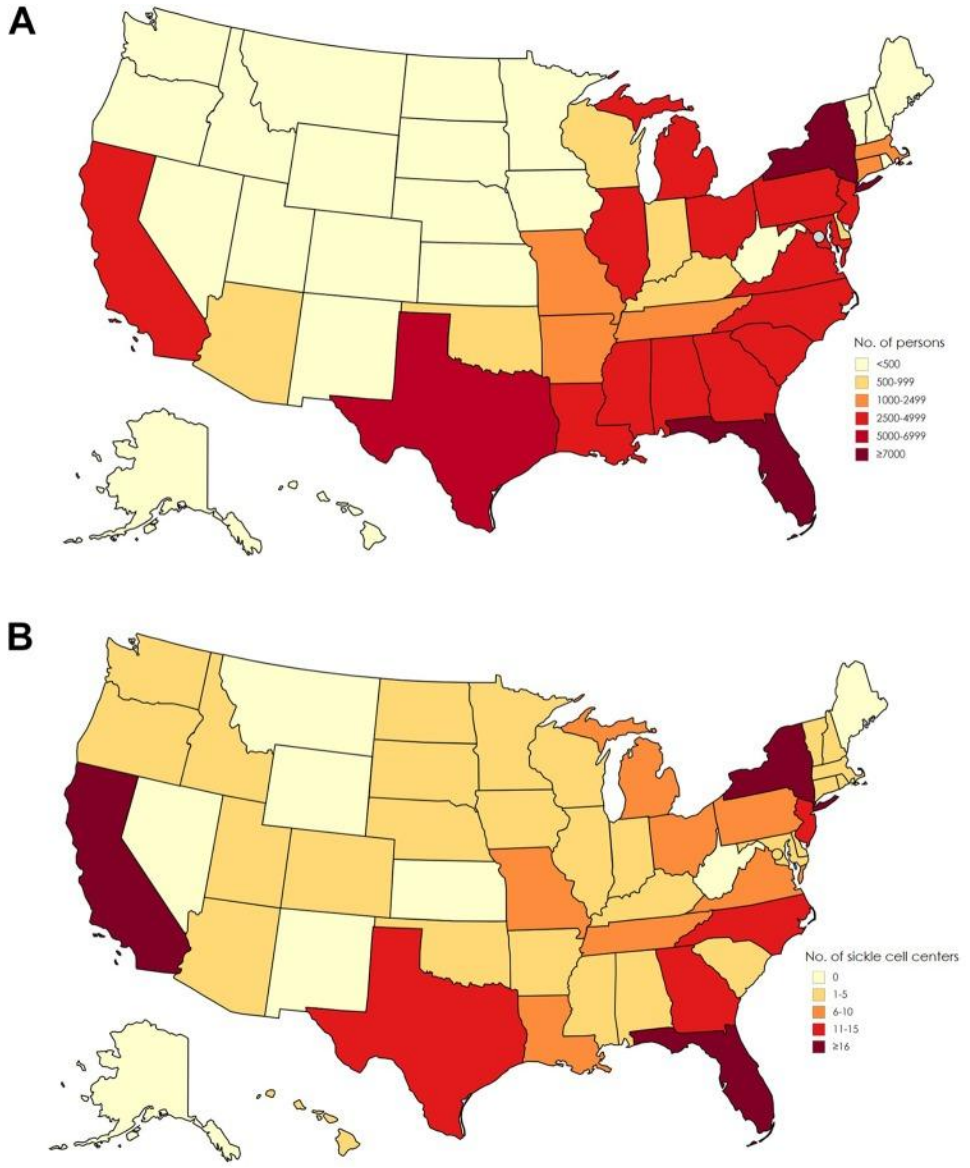
What is sickle cell disease (SCD)?

- Genetic blood disorder that causes some blood cells to be stiff and sickle-shaped. These blood cells can clump together, preventing oxygen flow through the body.[1]
- Major symptoms include chronic and acute pain (including severe pain episodes that require ED/hospitalization), fatigue, strokes, and organ disease.[2]
- People in the US with SCD live, on average, 20 years shorter than people without the disease.[2]



SCD prevalence in the US

- Approx 100,000 people in the US living with SCD[2]
- >90% of people in the US with SCD are Black or African American, 3-9% are Hispanic or Latino[2]
- Large disparities in access to care: there are fewer SCD treatment centers and less money is invested in R&D for SCD relative to other, even rarer diseases such as cystic fibrosis and hemophilia[3]



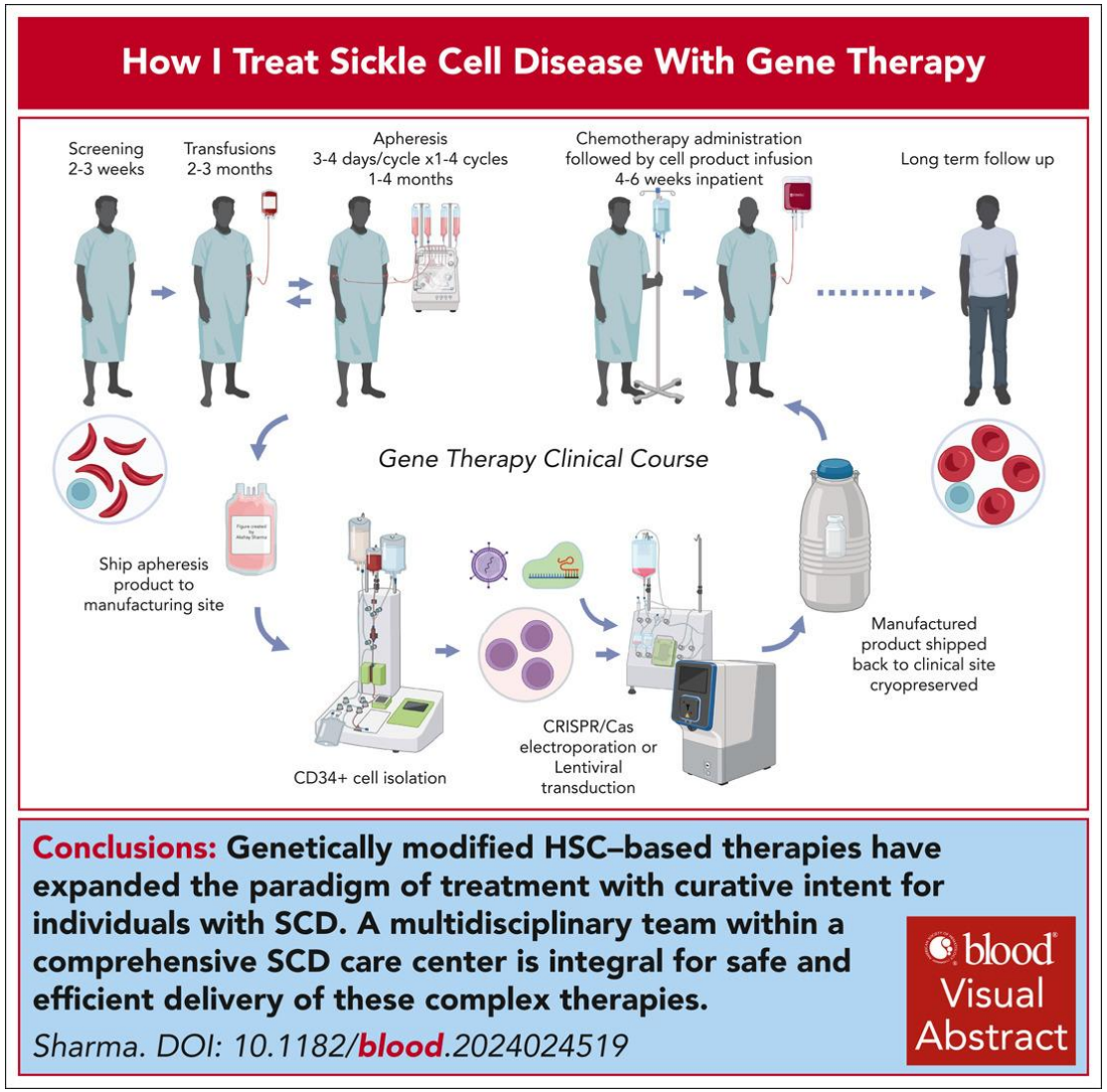
A) Estimated number of persons with sickle cell disease, based on state-specific African American and Hispanic birth-cohort disease prevalence and 2008 US Census population, corrected for early mortality. Map previously published by Hassell[1] and reprinted with permission from the American College of Preventive Medicine. (B) Number of sickle cell disease treatment centers per state, 2017. Data source: Centers for Disease Control and Prevention (CDC).[17] Used with permission from CDC.

Image source: (3) Lee et al. 2019, Public Health Rep. & Haskell

Two gene therapies (GT) for SCD have been approved in the US[1]

GT name	Manufacturer	Approved for[4,5]	How it works	List price[6]
Lyfgenia	bluebird bio	Patients 12+ years old with SCD and history of vaso-occlusive events (VOEs)	Inserts a working hemoglobin gene into patients' bone marrow cells	**$3.1 million**
Casgevy	Vertex Pharmaceuticals	Patients 2+ years old with SCD, recurrent vaso-occlusive crises (VOCs) and transfusion-dependent ß-thalassemia (TDT)	Edits patient's own gene so they produce working hemoglobin	**$2.2 million**

How does gene therapy for SCD work?



- Patients own cells are collected, modified, and re-administered
- Significant inpatient care and support services required
- Chemotherapy regimen impacts patients' fertility

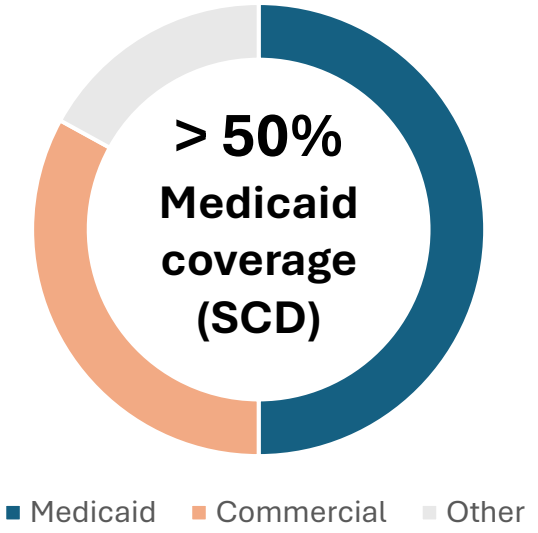
- **Economics note**: Because patient's own cells are modified, GTs have high marginal costs

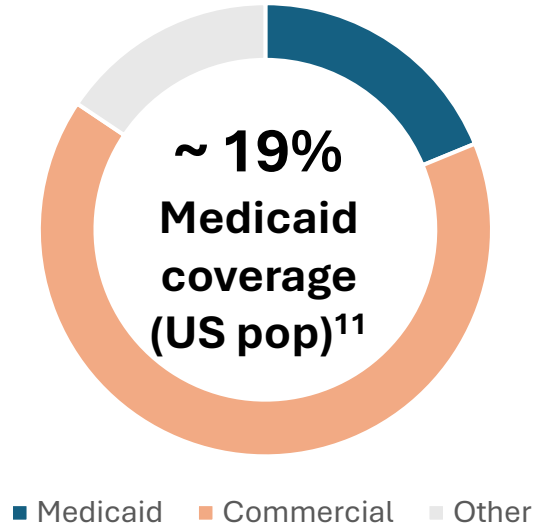
Source: (7) Sharma A. How I treat sickle cell disease with gene therapy. *Blood*. 2024;144(26):2693-2705. doi:10.1182/blood.2024024519

Sickle cell costs in the US: current state

- Of the ~100,000 people with sickle cell in the US
  - Over 50% have health insurance coverage through Medicare or CHIP (Children's Health Insurance Program)[8]
  - 1/3 have commercial insurance coverage[10]

- Estimated lifetime costs for a person with sickle cell: **$1.7 million per person**, with $44k paid out of pocket[9]





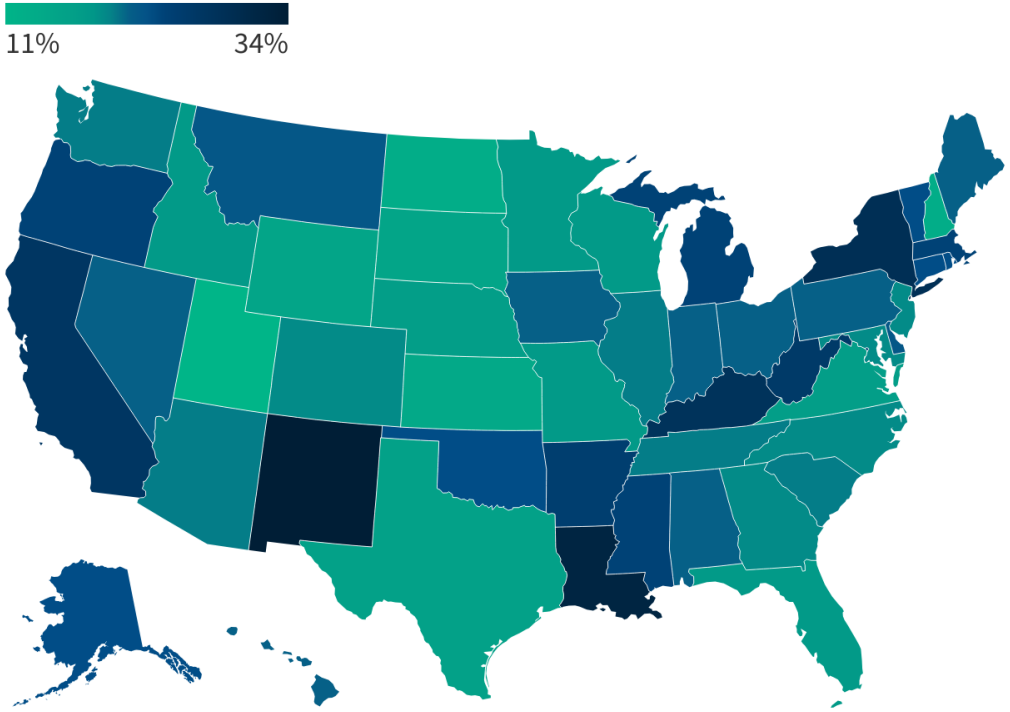
Medicaid provides health insurance for low-income Americans

- Medicaid is a program that provides health insurance to low-income people in the US
- Medicaid is funded by a combination of state and federal dollars[10]
- States have a lot of flexibility in deciding who to cover and which services to provide
- Individuals with SCD are disproportionately covered by Medicaid (~50%)[8]

Figure 2

Nationally, One in Five People Have Medicaid, but This Varies Across the States

Share of people enrolled in Medicaid by state



Note: This figures includes people who are covered by Medicaid-only as well as those who have both Medicaid and another type of coverage.

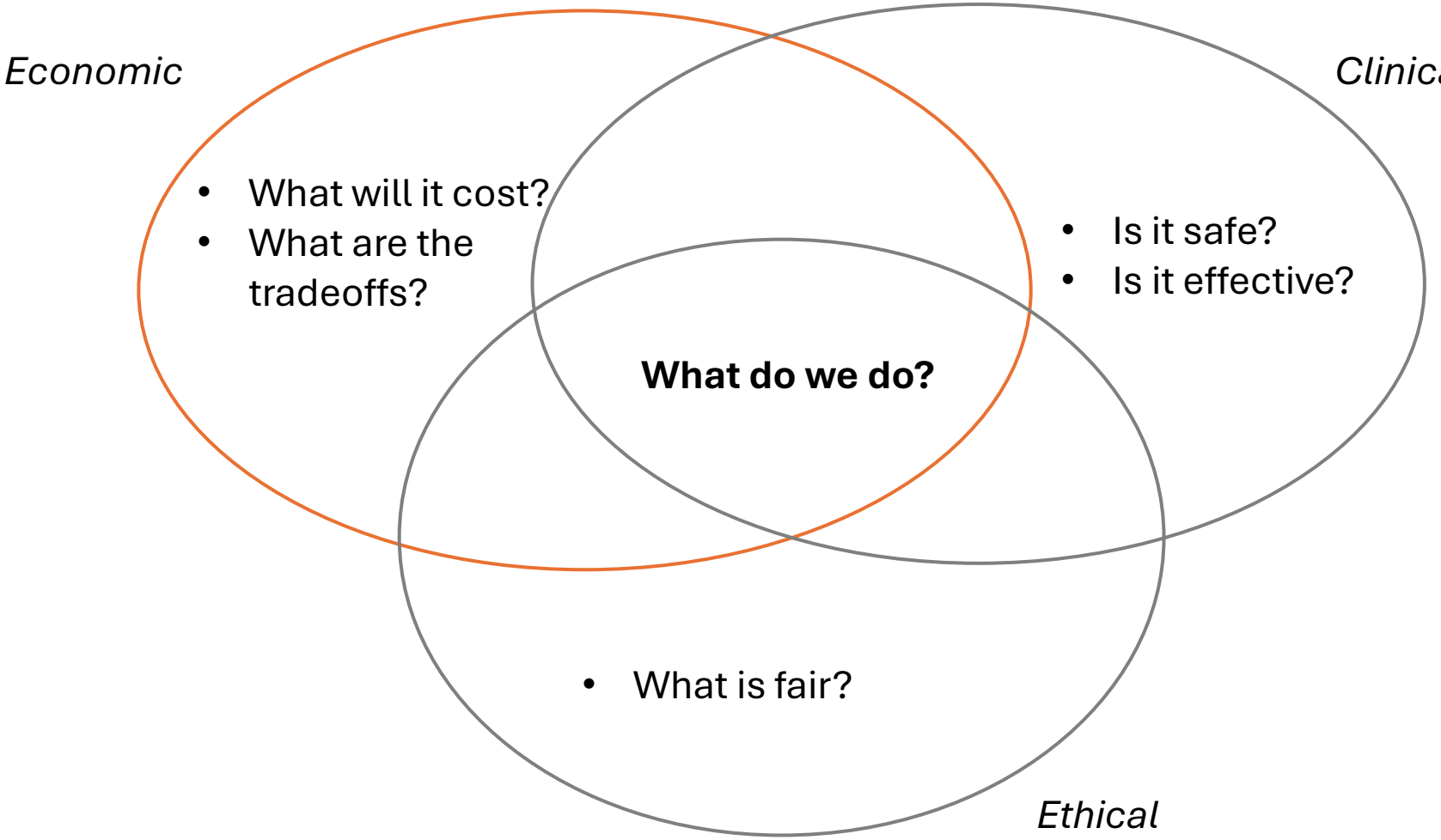
Source: KFF State Health Facts, Health Insurance Coverage of the Total Population, 2023

KFF

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Economics is only one of many lenses needed to evaluate questions prompted by gene therapies



Your task: Decide how gene therapies for SCD will be allocated to Medicaid enrollees in your state

- Who will be eligible to receive SCD GT?
  - Most generous: Anyone who meets FDA approval criteria can receive
  - Least generous: No one gets access
  - *Can pick any option along this spectrum*

- Who will be prioritized and why?
  - Each year, assume only 10 people max can access gene therapy for SCD in your state. How will you decide who gets priority?
  - *Example allocation strategies (non-exhaustive):*
    - First come, first served
    - Priority to sickest
    - Priority to youngest
    - Random lottery

Everyone will have 15-20 minutes to prepare their proposal, and 5 minutes to present it to the class and answer questions.

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Questions for discussion

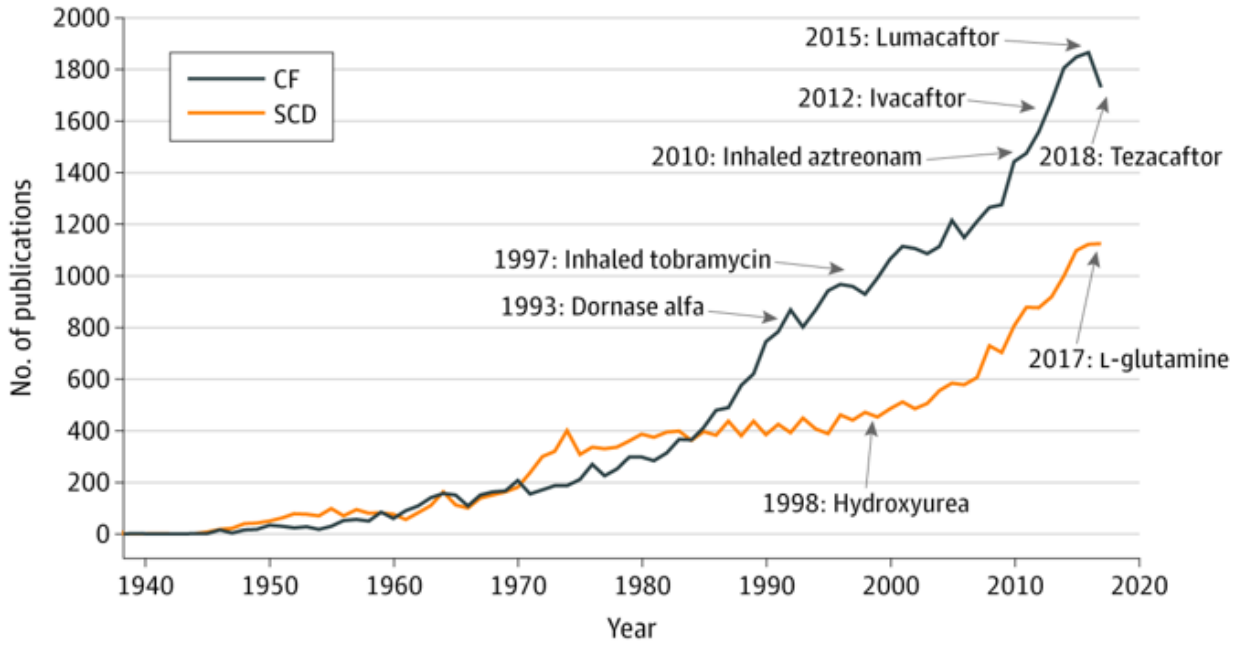
- Did cost-effectiveness factor into your decision to be more or less generous in your gene therapy coverage? Why or why not?
- What are the pros and cons of the different prioritization strategies groups proposed? Did any seem intuitively more or less appealing?
- Did you change your mind about which approach you would support after hearing another group's proposal?
- Was there any information you would have liked to have but couldn't find?

A note on health disparities: SCD and cystic fibrosis[12]

Table 1. Summary of Disease Characteristics, Funding, and Research Output

	SCD	CF	*P* value
Disease characteristics			
Patients, No.[19-22]	90 000	30 000	NA
US birth incidence			
White	1/123 000	1/2600	NA
Black	1/314	1/6000	NA
Hispanic[1-3]	1/16 300	1/9200	NA
Life span, mean, y[20,23]	58	46	NA
US mortality in 2015, No.[24]	903	540	NA
Estimated lifetime costs per individual, $[25,26]	460 151	306 332	NA
Annual funding (2008-2017)			
NIH funding (in millions), mean (SD), $	76.3 (13)	84.2 (5.2)	.05
NIH funding per person affected, mean (SD), $	812 (147)	2807 (175)	<.001
Foundation expenditure (in millions), mean (SD), $	9.14 (1.2)	231 (119)	<.001
Foundation expenditure per person affected, mean (SD), $	102 (13.7)	7690 (3974)	<.001
Total funding per person affected, mean (SD), $	943 (148)	10 592 (3841)	<.001
Annual NIH career awards, mean (SD), No.	16.7 (2.87)	16.6 (1.74)	.92
Research output (2008-2018)			
Annual PubMed publications, mean (SD), No.	926 (157)	1594 (225)	<.001
Annual clinical trials, mean (SD), No.	24 (6.3)	27 (6.9)	.23
New FDA drug approvals, No.	1	4	NA
Novel FDA drug indications, No.	2	11	NA

Figure 1. Number of Disease-Specific PubMed Listings and US Food and Drug Administration Drug Approvals Over Time



Abbreviations: CF, cystic fibrosis; FDA, US Food and Drug Administration; NA, not applicable; NIH, National Institutes of Health; SCD, sickle cell disease.

From: **Comparison of US Federal and Foundation Funding of Research for Sickle Cell Disease and Cystic Fibrosis and Factors Associated With Research Productivity**

JAMA Netw Open. 2020;3(3):e201737. doi:10.1001/jamanetworkopen.2020.1737

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SCD affects people worldwide – can/should we expand gene therapy access worldwide?

Global incidence of sickle cell anemia (severe sub-type of SCD)

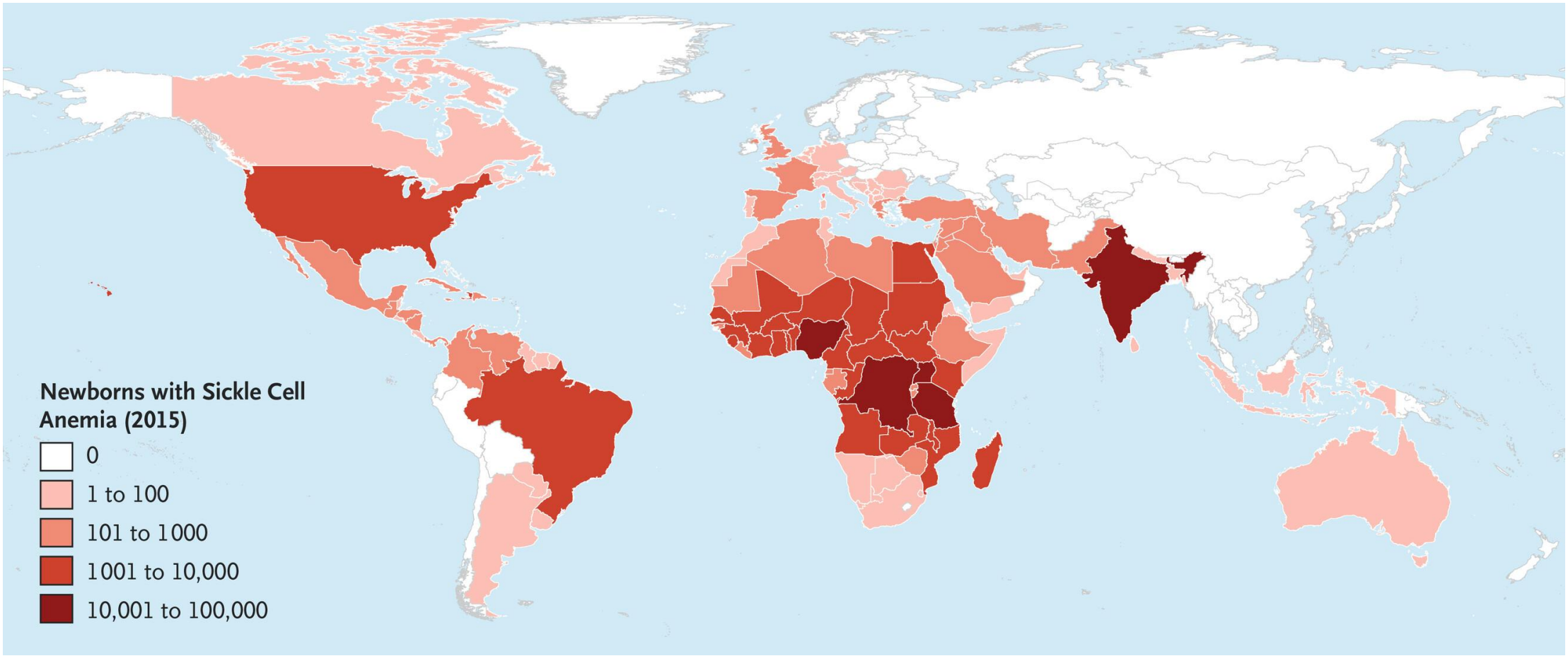


Image source: (13) Piel et al. 2017, NEJM

Could any of the strategies from Tuesday's video help get SCD treatments to LMICs?

1. Not-for-profit research
2. Advance market commitments
3. Patent buyout
4. Priority review vouchers

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Maybe, but still need to consider:

- GTs have **high marginal costs**. Even if we get prices for LMICs down to MC, they may still be unaffordable.
- Lots of **high-intensity support services needed** for GT (chemo, hospital stay, SCD specialists, etc.). These would also need to be built up in LMICs (and in lower resourced areas of the US!) for people to have meaningful access to GTs.

Upcoming

- Thursday, Oct 15: meet with Unit 2 project team in lecture
- Friday, October 23: Unit 2 assignment due

Sources

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